

RBG-48294

10 CFR 50.73

June 4, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2023-005-01, Manual Reactor Scram Due to Lowering Feedwater Temperature Following an Automatic Isolation of Feedwater Heater String

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,



Randy Crawford

RC/db

Enclosure: Licensee Event Report 50-458 / 2023-005-01, Manual Reactor Scram Due to Lowering Feedwater Temperature Following an Automatic Isolation of Feedwater Heater String

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

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**Licensee Event Report 50-458 / 2023-005-01, Manual Reactor Scram Due to Lowering
Feedwater Temperature Following an Automatic Isolation of Feedwater Heater String**



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name River Bend Station, Unit 1	☒ 050	2. Docket Number 00458	3. Page 1 OF 3
	☐ 052		

4. Title Manual Reactor Scram Due to Lowering Feedwater Temperature Following an Automatic Isolation of Feedwater Heater String
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
11	17	2023	2023	- 005 -	01	06	04	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 024
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Randy Crawford, Regulatory Assurance Manager	Phone Number (Include area code) 225-381-4177
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	SJ	N/A	N/A	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On November 17, 2023, at 22:15 CST, with River Bend Station, Unit 1, operating in Mode 1 at 30% power, an isolation of Low-Pressure Feedwater Heater String 'A' occurred. Operators entered the applicable Abnormal Operating Procedure (AOP) and reduced power to mitigate the lowering feedwater temperature. Reactor power was lowered to 24% and a manual reactor scram was inserted at 23:55 CST. All control rods fully inserted and there were no complications. All systems responded as designed.

A direct cause of the Heater String isolation cannot be determined.

This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of the Reactor Protection System.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME River Bend Station, Unit 1	☒ 050	2. DOCKET NUMBER 00458	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER 005	REV NO. 01

NARRATIVE**Event Description**

On November 17, 2023, at 22:15 CST, River Bend Station (RBS), Unit 1, was operating at 30% power while performing plant startup activities when an isolation of Low-Pressure Feedwater Heater String 'A' [SJ] occurred. Main Control Room Operators entered Abnormal Operating Procedure – Loss of Feedwater Heating (AOP-0007) for transients involving lowering feedwater temperature. AOP-0007, Attachment 1 (Feedwater Temperature vs Core Thermal Power Graph), provides guidance for power operations with reduced feedwater temperature from 100% reactor power to 25% reactor power. In accordance with AOP-0007, Attachment 1, operators began to manually insert control rods using the Control Rod Drive System [AA] to exit the restricted region for power operations at the lower feedwater temperature. Control rod insertion lowered power to 24%, which placed the unit below the graph start point of AOP-0007, Attachment 1. At 23:55 CST, while RBS was operating at 24% power, a manual reactor scram was inserted due to operation below the graph start point of AOP-0007, Attachment 1, caused by continued lowering of feedwater temperature. All control rods fully inserted and there were no complications. All systems responded as designed. AOP-0007, Attachment 1, contains a graph of feedwater temperature versus core thermal power; however, no information is provided for power levels below 25.7% of rated power. With reactor power lowering to 24%, the operators were not able to determine if the unit was operating in an acceptable region. Therefore, the conservative decision was made to manually scram the reactor. Following the initial investigation, engineering input from calculations performed as part of an engineering change has provided new information for operating at power levels below 25.7%. This event was reported under 10 CFR 50.72(b)(2)(iv)(B), as any event or condition that results in actuation of the Reactor Protection System (RPS) when the reactor is critical and 10 CFR 50.72(b)(3)(iv)(A) Specified System Actuation as result of Group 3 isolations (EN 56863). This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of the Reactor Protection System.

Event Cause

There is insufficient evidence to determine a direct causal factor for the Low-Pressure Feedwater Heater String isolation event. An independent third-party review of this incident has determined the most probable cause of the isolation was a level switch actuation due to steam backflow from the drain receiver tank volume flashing as a result of a rapid pressure transient event.

Safety Assessment

The actual consequence was the initiation of a manual scram. Following the scram, reactor pressure was maintained by the Turbine Steam Drains [TF] and reactor water level was maintained by the Feedwater [SJ] system. The RBS power to flow map continued to meet the licensing acceptance criteria parameters. There were no actual consequences to general safety of the public, nuclear safety, industrial safety, and radiological safety for this event.

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YEAR 2023	SEQUENTIAL NUMBER 005	REV NO. 01		

NARRATIVE**Corrective Actions**

1. Clarifying information and technical basis was added to AOP-0007 to help operators better understand the applicable guidance for power operations at reduced feedwater temperatures.
2. The Feedwater Temperature vs Core Thermal Power Graph in AOP-0007 has been updated for power operations below 25.7%.
3. Installation of a time delay in the level sensing circuit that will aid in diagnosing a valid level instead of a pressure transient to prevent future occurrences of heater string isolation.

Previous Similar Events

A similar event occurred at RBS in 1986. As documented in LER-1986-001, a reactor scram resulted from a Low-Pressure Feedwater Heater String isolation due to a false high level indication. The false high level was caused by flashing in the level transmitter pots. A modification was made to the system which relocated the level switches to a leg of downward sloped pipe. The modification prevents water from accumulating that would result in steam flashes during power ascension. Although this is a similar event, the failure mechanism in LER-1986-001 is different from the one as described in this LER.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).